

Fundamentals of Interaction

Lecture 2

Outline

- Four approaches to Interaction Design
- Fundamental Principles of Interaction
- Seven stages of action
- Human Errors
- Brainstorming with FigJam

Four Approaches to IxD

- Human-Centered Design (HCD)
 - Users guide the design process, designers translate user's needs and goals into products
- Activity-Centered Design
 - Designers focus on the tasks and activities that need to be accomplished
- Systems Design
 - Designers focus on the components of a system
- Genius Design
 - Designers use their knowledge in human psychology and physiology to make products

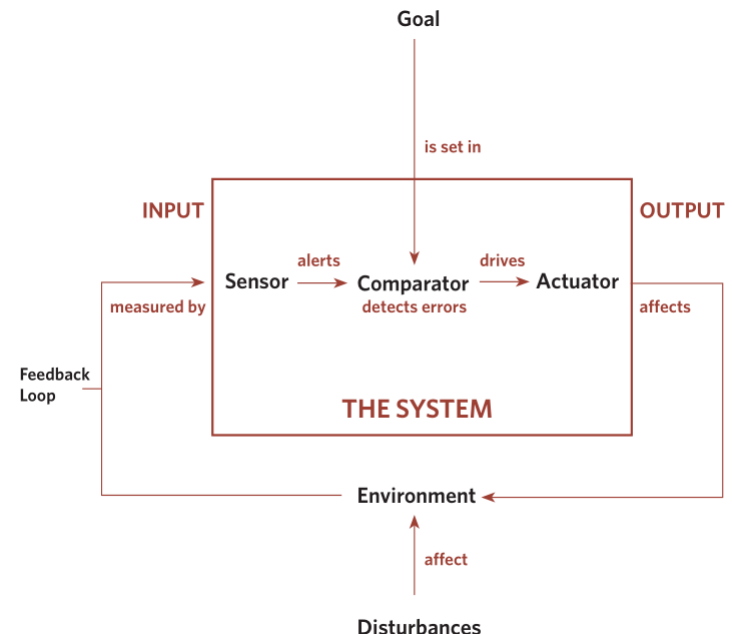
Activity-Centered Design

- ACD is used for functional tools such as appliances and cars
- More focused and tangible purposes
- Activities consist of discrete tasks
- Tasks require corresponding interface



Systems Design

- System, a set of entities that act upon each other is at the centre
- Offers structured, rigorous and holistic approach to complex problems
- Components of the system:
 - Goal
 - Sensor
 - Comparator
 - Actuator
 - Environment



Genius Design

- Designers use their best judgement as what users want
- Users are usually involved at the end of the process
- Mostly taken by experienced designers

the
Henry
Ford



Human-Centred Design

- An approach that puts human needs, capabilities, and behaviour first, then designs to accommodate those needs, capabilities and ways of behaving.
- Good design is about communication:
 - What actions are possible?
 - What is happening?
 - What is about to happen?



Darn these hooves!
I hit the wrong switch again!
Who designs these instrument
Panels, raccoon?!

Fundamental Principles of Interaction

- Affordances
- Signifiers
- Mapping
- Feedback
- Constraints
- Conceptual models

Affordance and Signifier

- Affordance

- Affordances determine *what actions are possible* between an object and an agent.
- Some affordances are perceivable, others are not.
- It is a relationship between the properties of an object and the capabilities of the agent that determine just how the object could possibly be used.

- Signifier

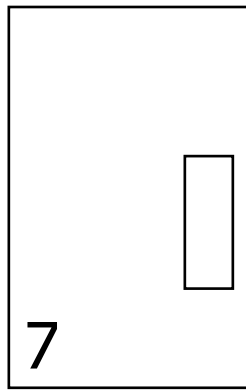
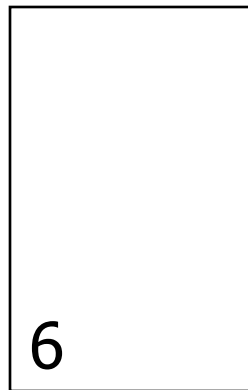
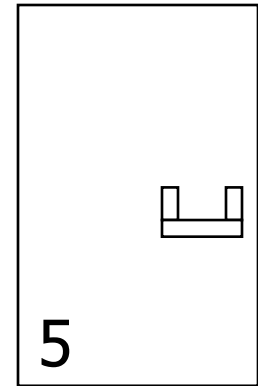
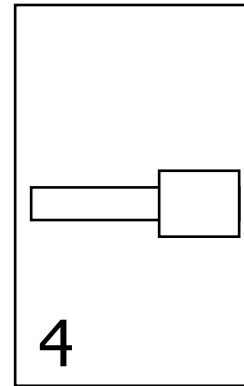
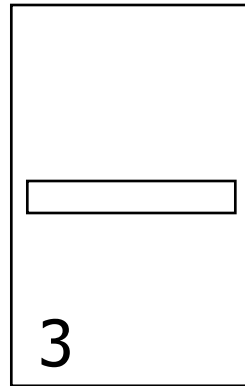
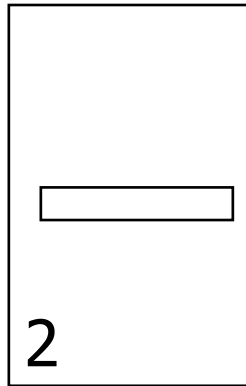
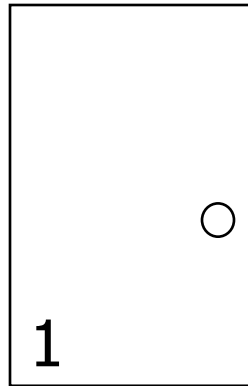
- Signifiers determine *where the action should take place* between an object and an agent.
- It refers to any mark or sound, any perceivable indicator that communicates appropriate behavior to an agent.

Affordance Examples

- Chair can support human weight
- Chair can be lifted by humans
- Chair can block the passage
- Chair can hold clothes
- Screen can be touched
- Screen can be swiped
- Screen can be turned
- Screen can lighten the space



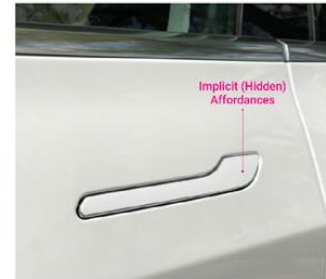
Door Opening Affordances



- Which doors are easy to open?
- Which doors are hard to open?
- Why?

Affordances and Signifiers

A.



B.



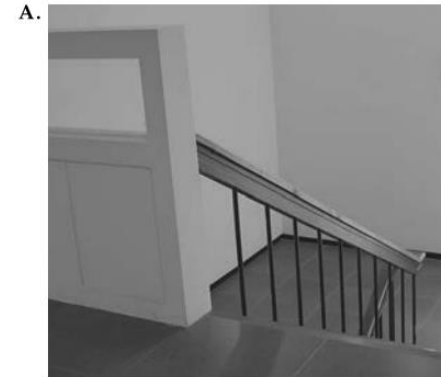
C.



Accidental Affordances



Men's bathroom in restaurant



Hidden Affordances

- Interfaces are virtual and do not have 'real' affordances like physical objects
- Some affordances are not visible, hence signifiers are used. Ex:
 - Swiping left or right brings up new restaurant recommendations
 - Swiping up reveals the menu for the restaurant being displayed
 - Swiping down, friends who recommend the restaurant



Anti-affordances



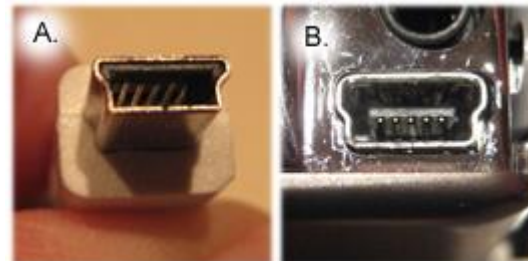
Norman Mantra

- Complex things may need signifiers, but simple things should not
- If a simple thing requires a signifier, it is likely a failed design

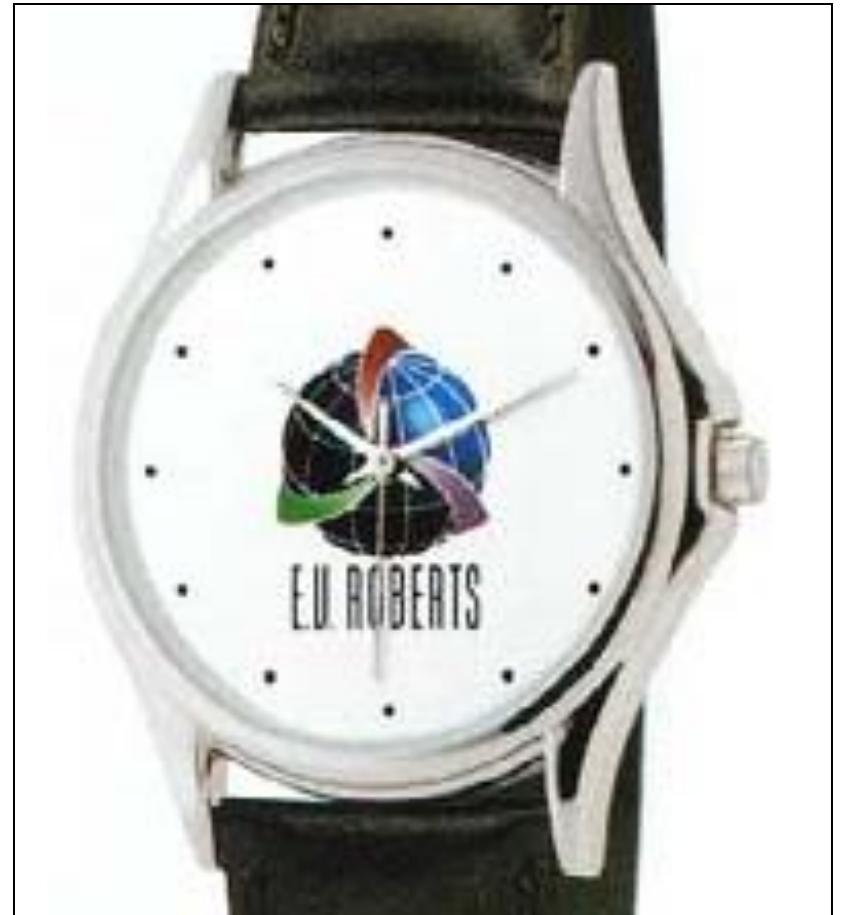
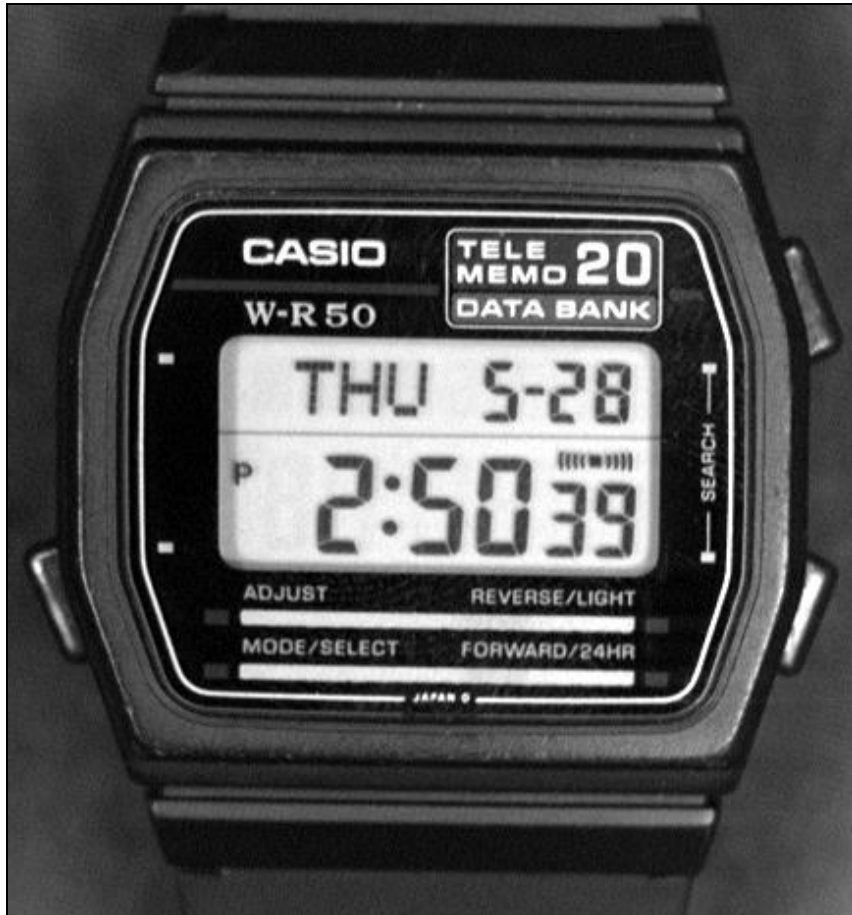
Signifier



Perceivable
Affordance



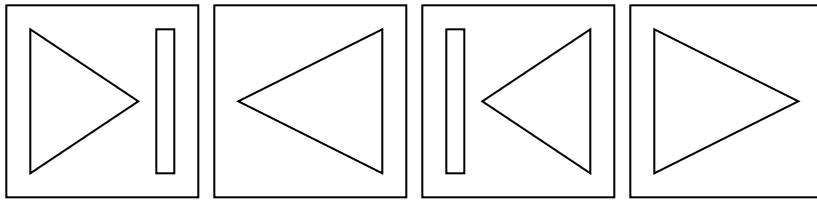
Which is Faster for Setting Time?



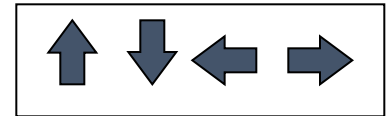
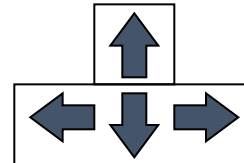
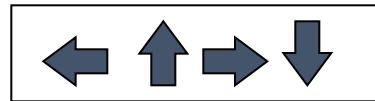
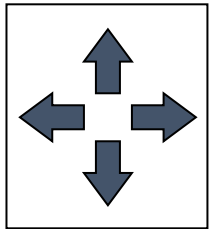
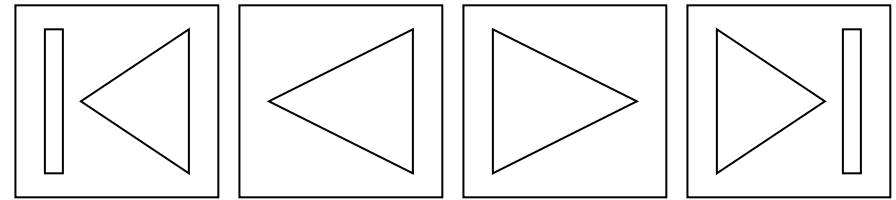
Mapping

- It is a relationship between **controls** the user interacts with and **functions** the device/application performs.
- When number of functions is greater than number of controls, some functionality is hidden to the user.
- When capabilities are visible and controls associated with them are correctly mapped, interaction is intuitive.
- Mapping can be realized in 3 ways:
 - Control is placed on the spot where function takes place
 - Control is placed near the spot where function takes place
 - Control shares natural similarity in shape, size and behaviour to a part which exhibits the function

Which is better?



or



Mapping Examples – car window

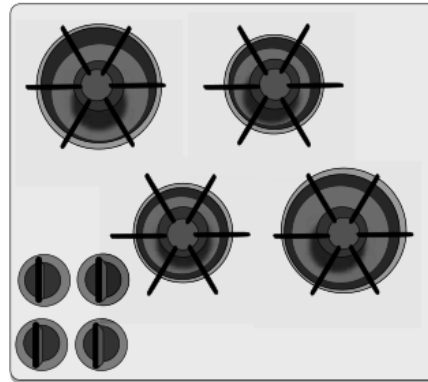
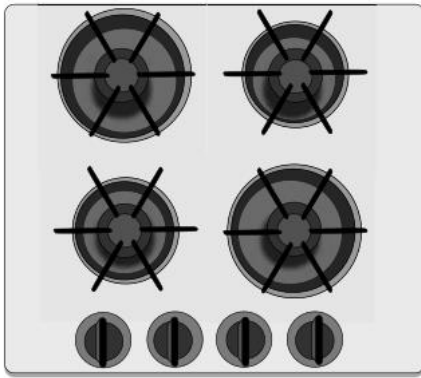


- They are located on the center console between the front seats.
- One problem with these controls is that they are not where you expect them to be.
- When a person wants to raise or lower a window they generally will look over by the window for the control, not between the front seats.
- By the way, there is something else wrong with these controls: "How do you raise the window?"

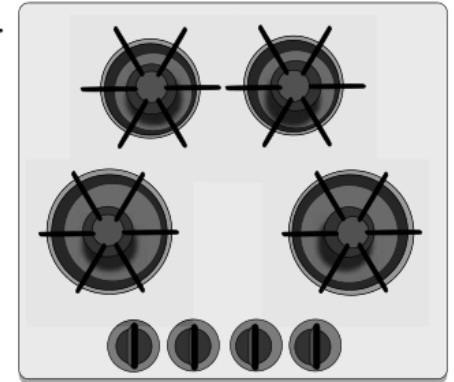
Mapping Example- Stove

Which controls which?

A.



D.



Mapping Example: Driver's Seat



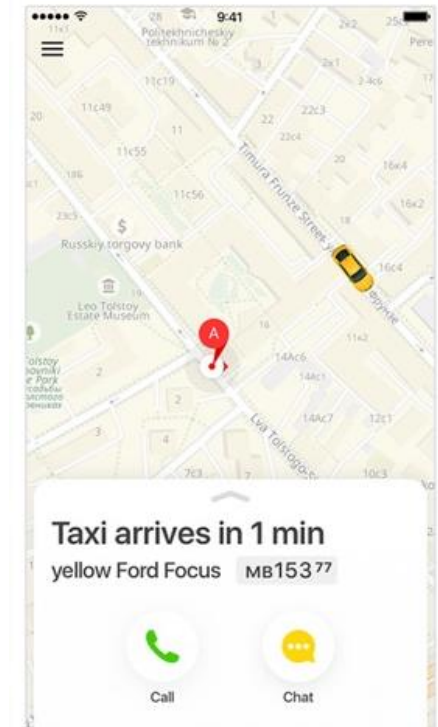
Feedback

- It is the way of communicating the results of an action to an agent.
- Feedback can be visual, auditory or any channel through which an agent can perceive information.
- Feedback must be immediate, informative, unobtrusive and ranked according to its level of alert.

Bad Feedback Examples



Good Feedback Examples



Constraints

- Restricting the wrong kind of interactions to make the correct ones more discoverable.
 - **Physical Constraints** prevent users from performing certain actions.
 - **Cultural Constraints** arise from the social and cultural context in which a product or system is used.
 - **Semantic Constraints** arise from the meaning of words and symbols used in a product or system.
 - **Logical Constraints** arise from the logical structure of a product or system.

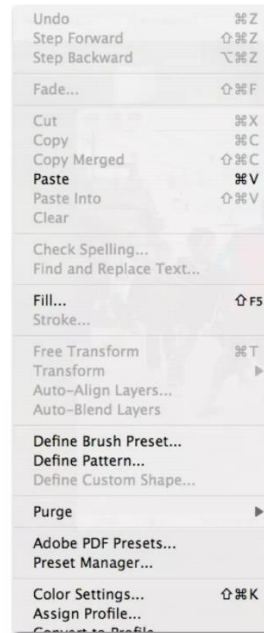
Constraint Implementations

- **Forcing Functions** – a failure at one stage prevents the next step from happening
- **Interlocks** – force operations to take place in a proper sequence
- **Lock-ins** – keep an operation active, preventing someone prematurely stopping it.
- **Lock-outs** – keep an operation passive, preventing someone intervening it (usually for safety reasons)

Constraint Example



Physical constraints



Semantic constraint



Cultural constraint

Sign Up

Username:

Username must be between 3 and 25 characters.

Email:

Password:

Password must have at least 8 characters that include at least 1 lowercase character, 1 uppercase character, 1 number, and 1 special character in (!@#%&^&*)

Confirm Password:

Please enter the password again

[SIGN UP](#)

Logical constraint

Good Constraint Example

A.



B.



Bad Constraint Example



Cylindrical Battery with Orientation



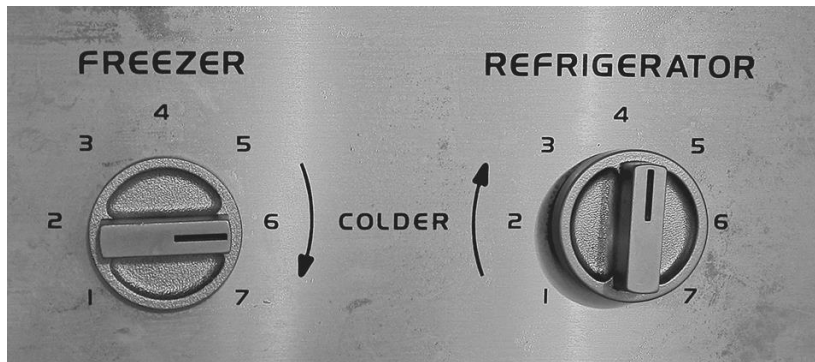
Cylindrical Battery without Orientation

Conceptual Models

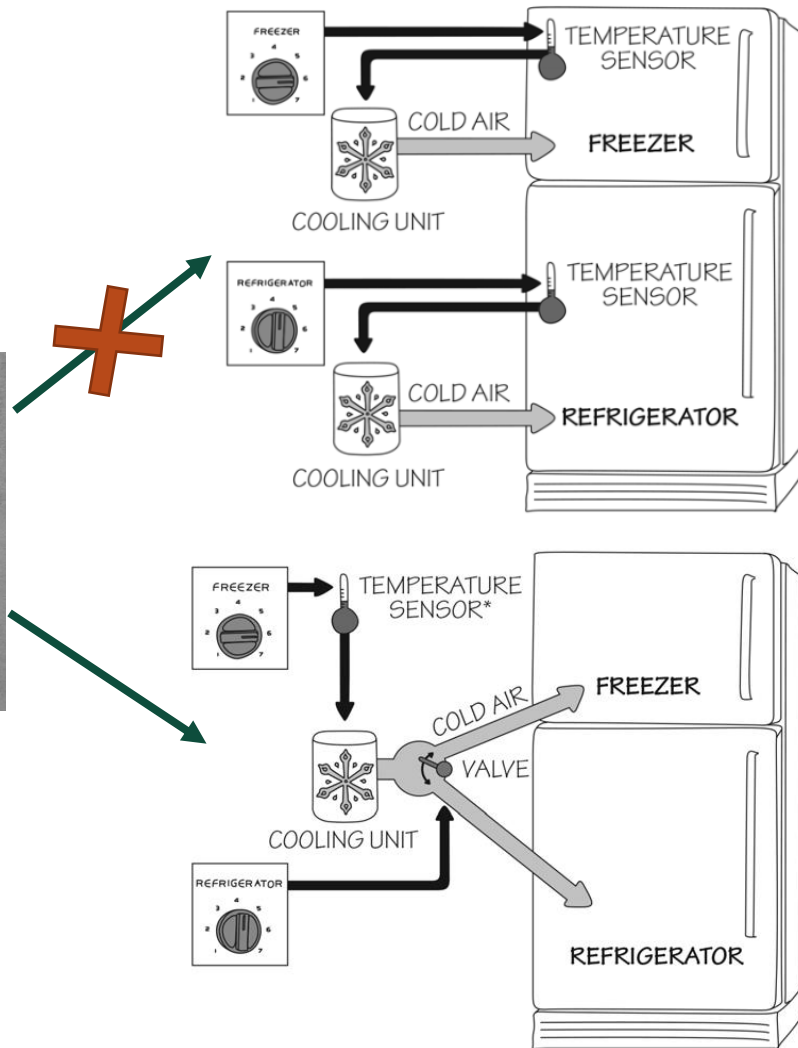
- Working principle of the device
 - User thinks it works one way
 - Designer knows exactly how it works
- Designer can help user foster an appropriate conceptual model
 - Appearance
 - Instructions
 - Behavior



Conceptual Model Example

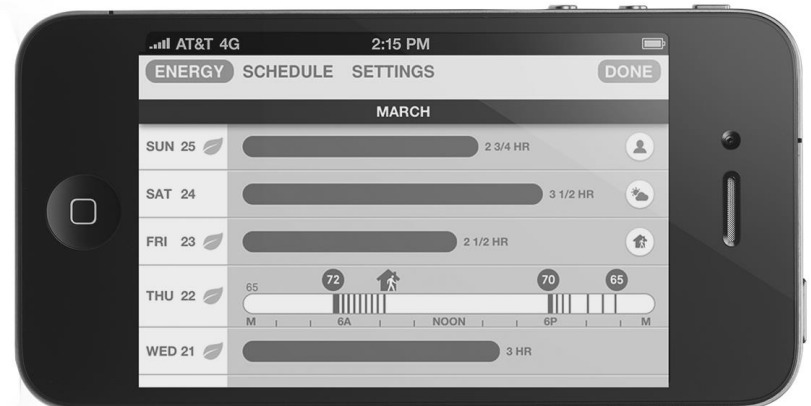


Refrigerator controls



* LOCATION UNKNOWN

Good Conceptual Model



A Thermostat with an Explicit Conceptual Model

Good Conceptual Model

Affordances - Insert something into holes

Constraints - Bigger hole for several fingers, small for thumb

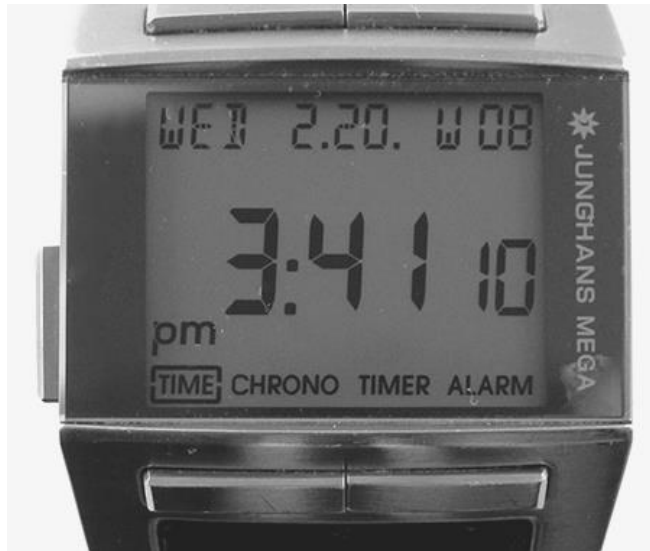
Mapping - How to insert fingers into holes suggested by visible appearance

Conceptual model - Suggested by how parts fit together and move

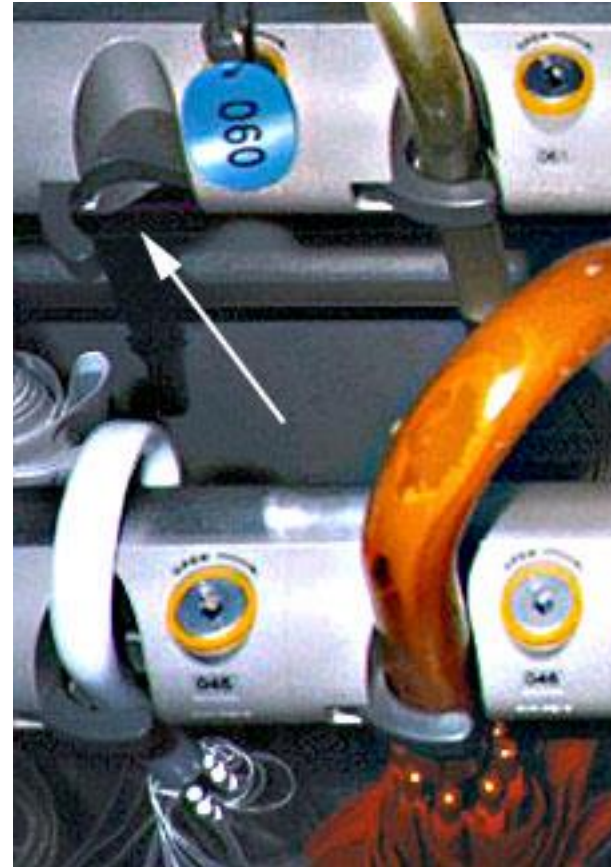
Scissors



Bad Conceptual Model



Digital watch with 5 buttons



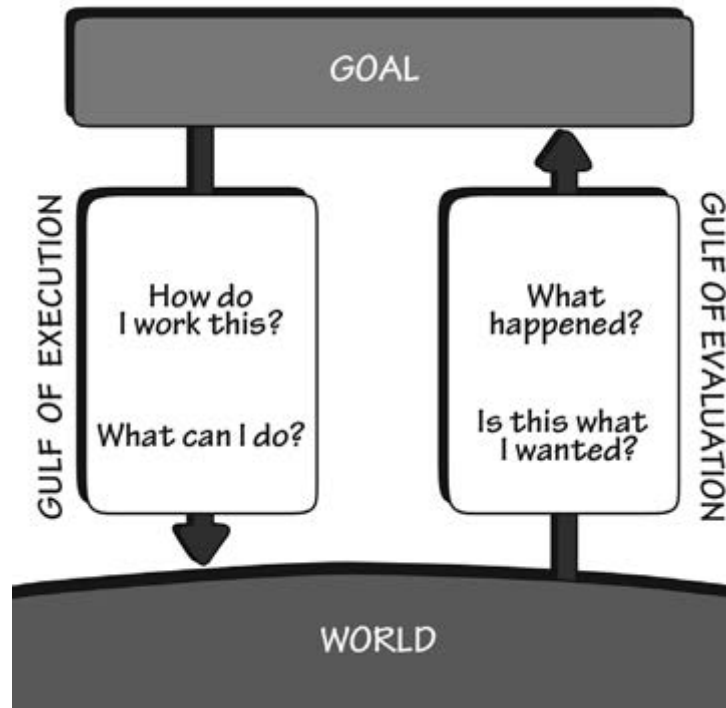
Umbrella hook at the entrance to museum in Tokyo

Recall vs. Recognize

- (Recall) Knowledge in Head
 - Not easy to retrieve
 - Learning required, good conceptual model makes easier
 - Can be very efficient
 - Not easy first time
 - Aesthetics do not need to make info visible
- (Recognize) Knowledge in World
 - Easy to retrieve
 - No learning, only interpretation
 - Use slowed by need to find the info to interpret
 - Easy for first time
 - Can be cluttered or inelegant

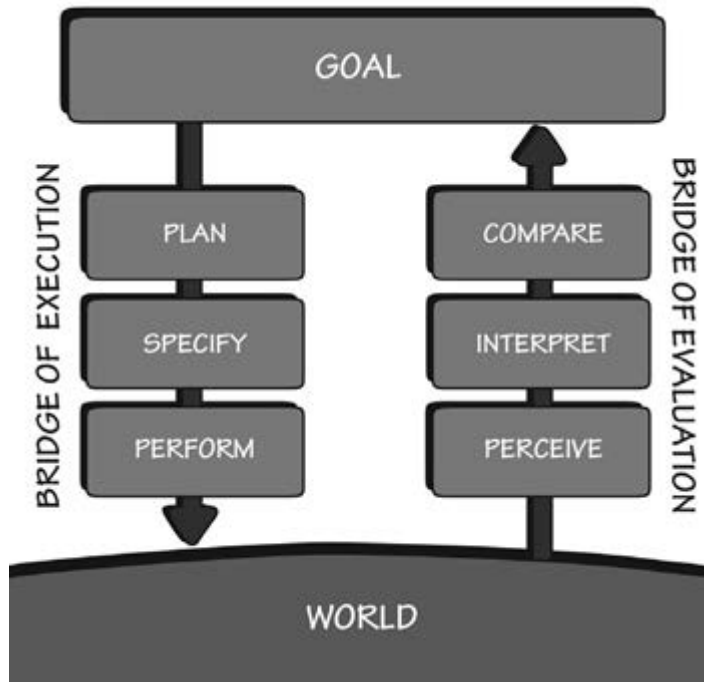
Goals, Execution, Evaluation

- Conceptual model
- Affordances
- Constraints
- Natural mappings



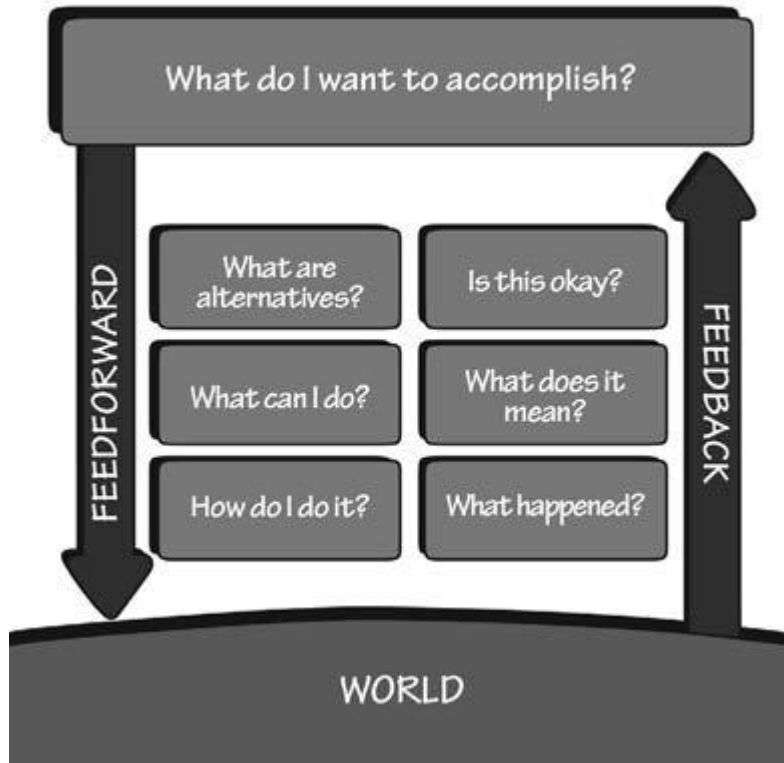
- Conceptual model
- Feedback

Seven Stages of Action



- User encounters a problem in a certain situation, and his **goal** could be solving this problem.
- For that he has to bridge 2 gulfs:
 - **Gulf of execution:**
 - **Plan:** what options do I have?
 - **Specify:** what are the steps?
 - **Perform:** how to perform these steps?
 - **Gulf of evaluation:**
 - **Perceive:** what has happened?
 - **Interpret:** what does it mean?
 - **Compare:** is it what I expected?

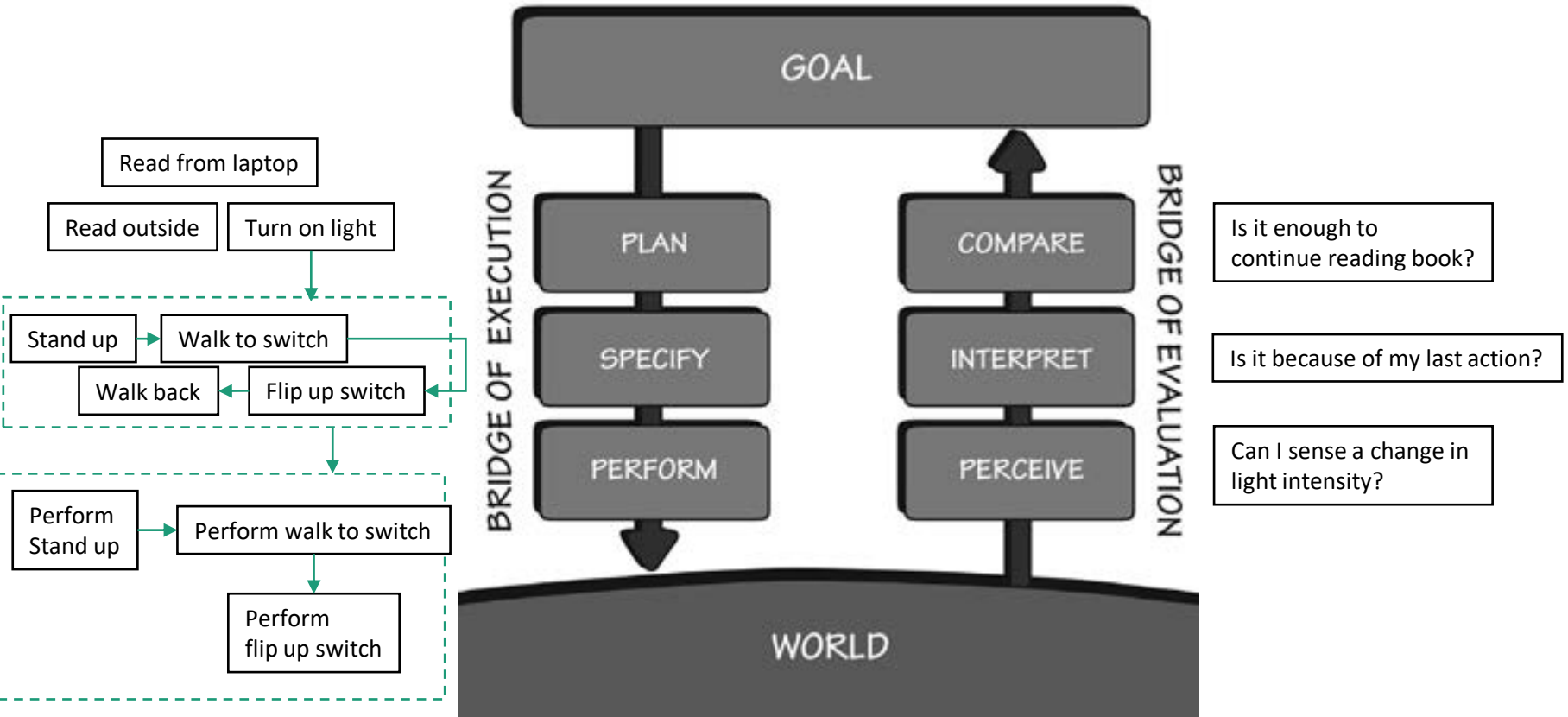
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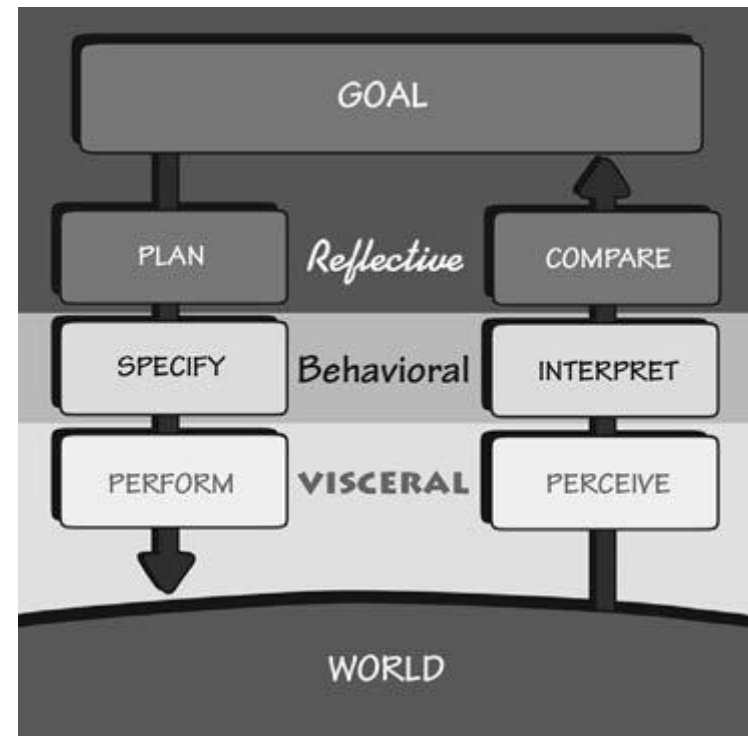
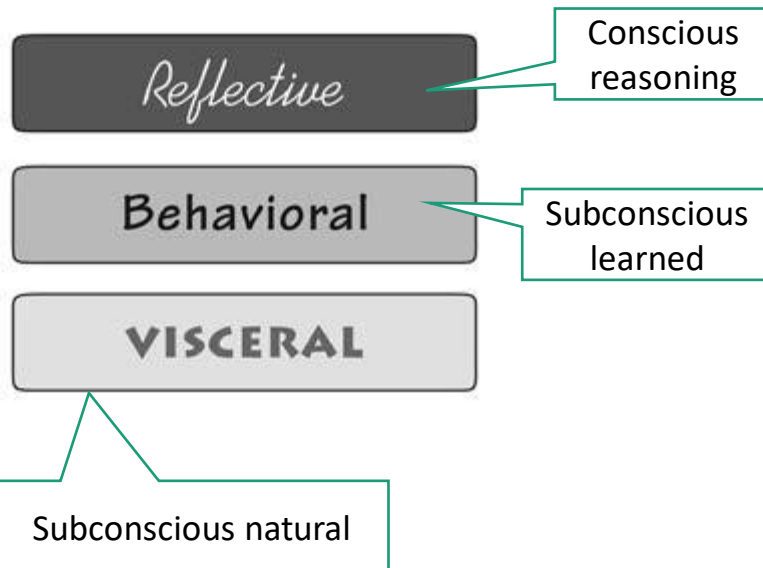
Example

It turns dark, I want to continue reading a book.



Three-levels of processing

Three Levels of Processing



How to bridge the gulfs?

Discoverability

- **Affordances**

- The proper affordances exist to make the desired actions possible.

- **Signifiers**

- Effective use of signifiers ensures discoverability and that the feedback is well communicated and intelligible.

- **Mappings**

- The relationship between controls and their actions follows the principles of good mapping, enhanced as much as possible through spatial layout and temporal contiguity.

- **Feedback**

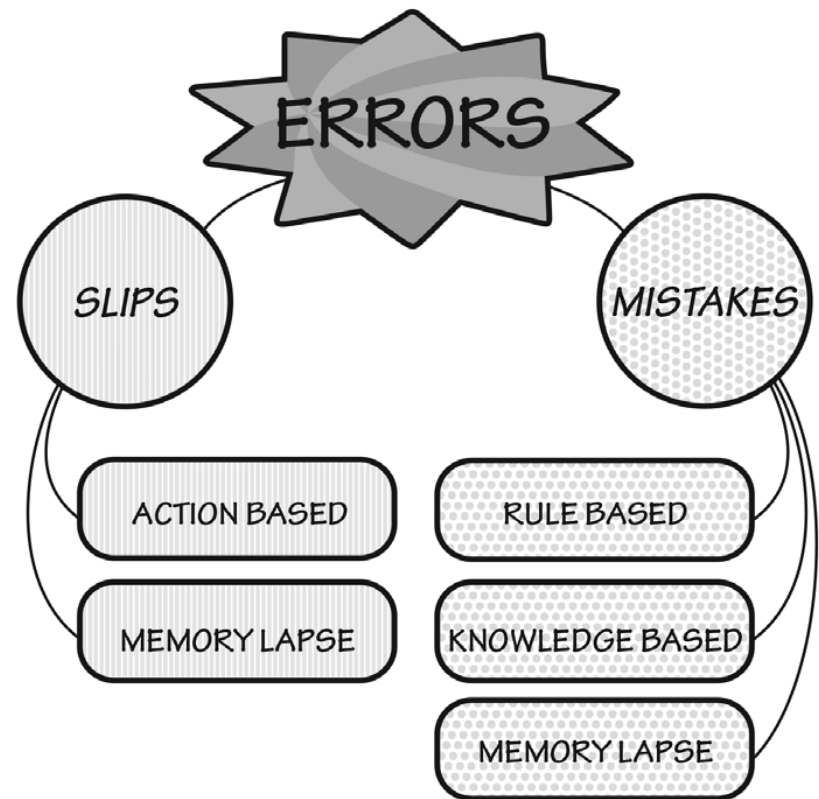
- There is full and continuous information about the results of actions and the current state of the product or service. After an action has been executed, it is easy to determine the new state.

- **Conceptual model**

- The design projects all the information needed to create a good conceptual model of the system, leading to understanding and a **feeling of control**.
- The conceptual model enhances both discoverability and evaluation of results.

Mistakes and Slips

- Mistakes (hopefully avoidable)
 - Wrong intention
 - Incorrect mental model
 - Novice behavior
 - Happen at conscious level
- Slips (unavoidable)
 - Wrong execution
 - Skilled behavior
 - Happen at subconscious level



Mistakes and Slips

- **Action-based Slip:**

- goal and plan are correct
- specification or execution are incorrect

- **Memory-lapse Slip:**

- **goal** -> **plan** transition happen
- **plan** -> **specify**, or **specify** -> **perform** transition does not happen
- **world** -> **perceive**, or **perceive** -> **interpret** transition does not happen

- **Rule-based Mistake:**

- goal is correct (or problem diagnosed properly)
- plan is incorrect, or
- interpret is incorrect

- **Knowledge-based Mistake:**

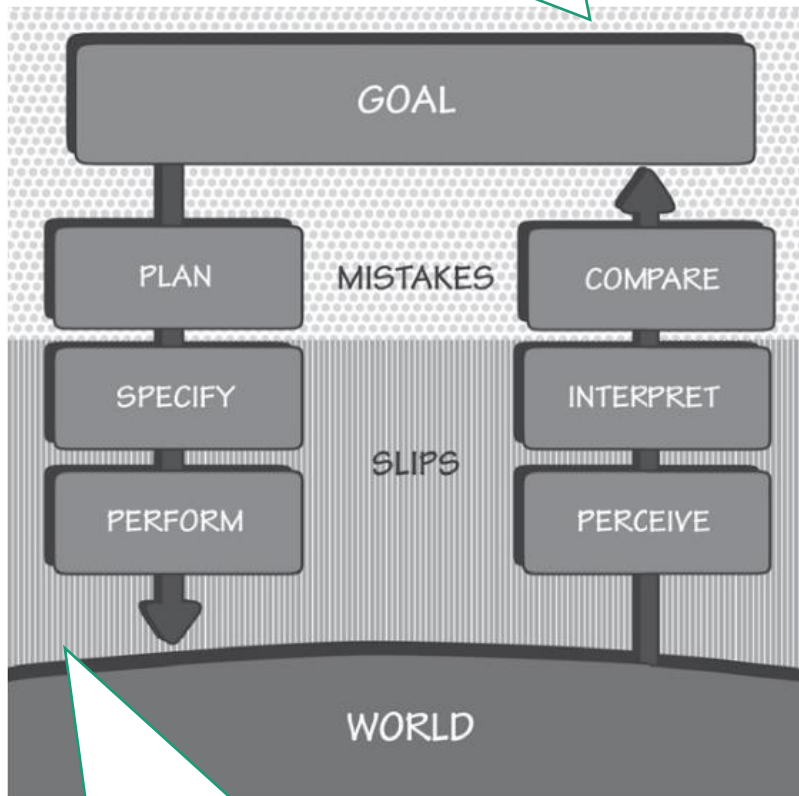
- goal is incorrect (or problem is misdiagnosed)
- compare is incorrect
- plan is correct/incorrect

- **Memory-lapse Mistake:**

- **goal**->**plan** transition does not happen
- **interpret**->**compare** transition does not happen
- **compare**->**goal** transition does not happen

Where slips and mistakes originate?

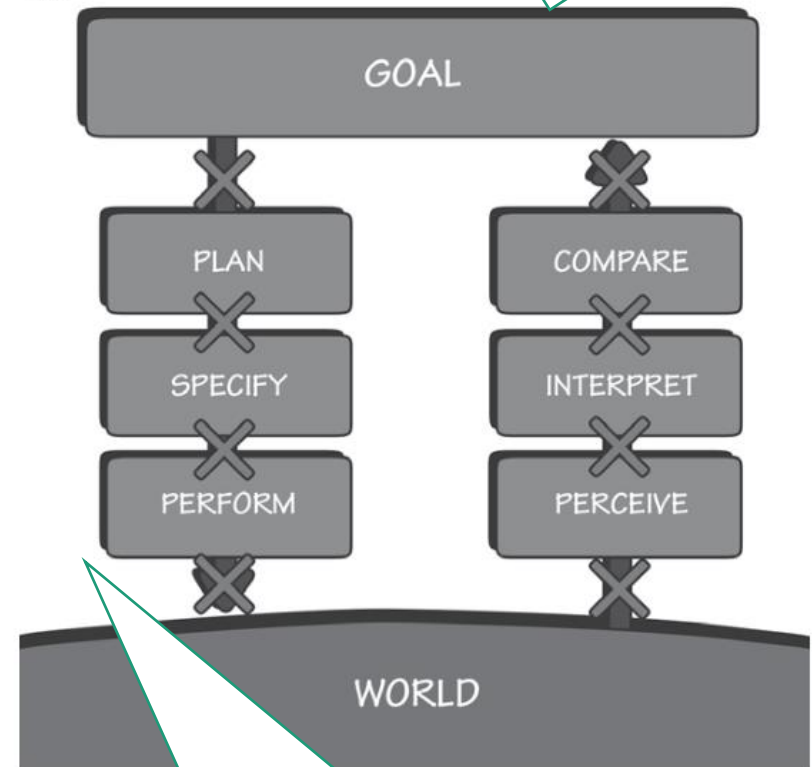
Rule-based or
Knowledge based mistakes



Action-based slips

Memory-lapse mistakes prevent
transition in higher levels

B.



Memory-lapse slips prevent
transition at lower levels

Examples

- Action-based slip
 - I poured some milk into my coffee and then put the coffee cup into the refrigerator.
 - This is the correct action applied to the wrong object.
- Memory-lapse slip
 - I forget to turn off the gas burner on my stove after cooking dinner.
- Rule-based mistake
 - I wanted to listen to my voice mail, but instead dialled police
- Knowledge-based mistake
 - Weight of fuel was computed in pounds instead of kilograms
- Memory-lapse mistake
 - A mechanic failed to complete troubleshooting because of distraction

Types of Slips

- **Capture error**

- Continue frequently done activity instead of intended one (similar starts)
- Example: Confirm deletion of file instead of cancel

- **Description error**

- Intended action has much in common with others possible (usually when distracted, close proximity)
- Example: Press CapsLock key, instead of Shift

- **Data driven error**

- Triggered by arrival of sensory info which intrudes into normal action
- Example: Call to give someone a number, dial that number instead

Types of Slips

- **Loss of activation**
 - Forgetting goal in middle of sequence of actions
 - Example: Start going into room, then forget why you're going there
- **Associative activation**
 - Internal thoughts and associations trigger action
 - Example: Phone rings, yell "come in"
- **Mode errors**
 - Do action in one mode thinking you're in another
 - Example: In Word, press Ctrl+T to create a new tab

Perceptual Errors

- Result from insufficient or poor perceptual cues
 - Display of objects that are visually similar
 - Invisible or poorly expressed states
 - Failure to capture user's attention
 - Lack of perceivable feedback

Cognitive Errors

- Caused by exercising the memory and problem solving capabilities
 - Recall memory
 - Lack of or poor mnemonic aids
 - Inconsistency
 - Lack of context or status info
 - e.g., where came from in a menu
 - Mental calculations and translations

Motor Errors

- Exercise the eye-hand coordination and motor skills
 - Awkward motor movements
 - Highly similar motor sequences
 - e.g., double click, click
 - Pressure for speed
 - Require a high degree of hand-eye coordination
 - Requiring special types of motor skills (type)

Errors - Three Considerations

- Avoiding and preventing
- Identifying and understanding
- Handling and recovering



Error Prevention Guidelines

- Eliminate modes or provide visible cues for modes - no invisible modes
- Maximize recognition, minimize recall
- Design non-similar motor sequences for commands
- Minimize need for typing
- Test and monitor for errors and engineer them out
- Allow reconsideration of action by user (e.g., removing file from trash)
- Avoid ambiguous and unclear prompts and messages - as in next set of examples!

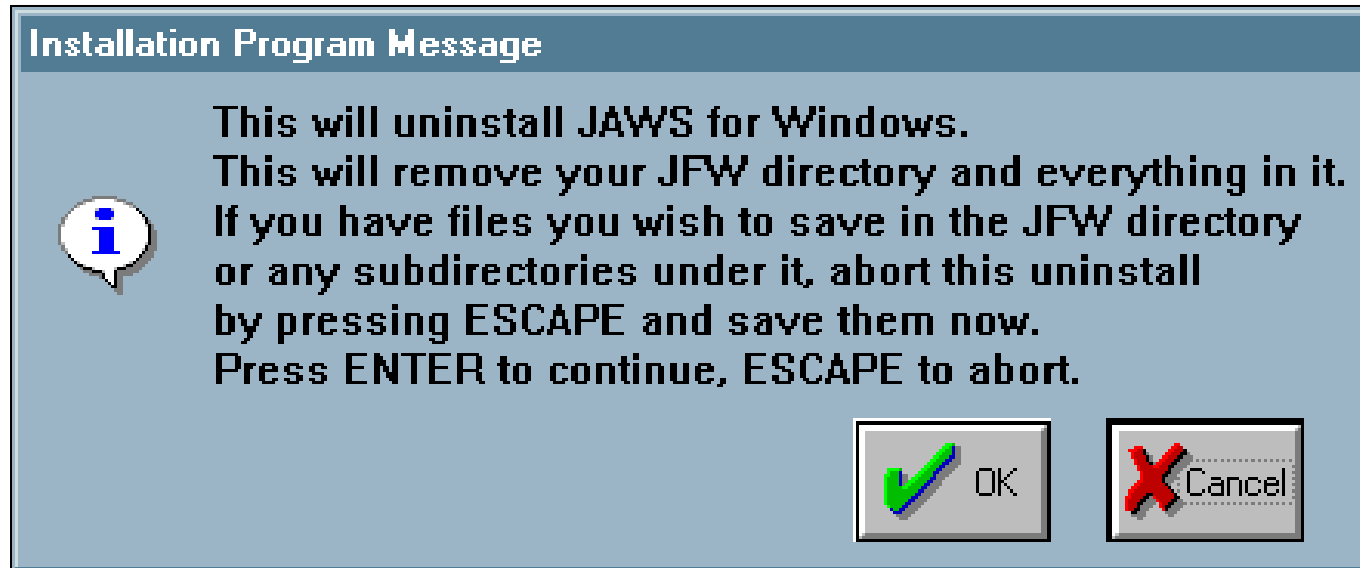
Error Recovery Guidelines

- Provide appropriate type of response
 - Gag - Prevent user from continuing
 - Erroneous login
 - Warn - Warn user an unusual situation is occurring
 - Bell or alert box
 - Nothing - Don't do anything (Careful, user must determine problem)
 - move file to bad place
 - Self-correct - Guess correct action & do it
 - Spell-check correction
 - Dialog - System opens dialog with user
 - Go into debugger on run-time crash

Error Recovery Guidelines

- Provide undo function
- Provide cancel function from operations in progress
- Require confirmation for drastic, destructive commands
- Provide reasonableness checks on input data
 - Did you really mean to order 5000?
- Return cursor to error field, allow fix
- Provide some intelligence
 - Guess what they wanted to do
- Provide quick access to context-sensitive help

Error Prevention/Recovery Examples



Norman summary

- Affordances are important
- Minimize the gulf of interpretation and gulf of execution
- Use natural mappings
- Make state visible
- Use a conceptual model that makes sense
- Provide feedback

Student Activities

using [FigJam](#)

Brainstorming Activity

- Think about a simple procedure of organizing students into teams.
- Suggest potential solutions for this problem.
- Analyze chosen solution as follows:
 - Determine affordances, signifiers, mappings, feedback, constraints and conceptual model
 - Determine the goal and put these elements into gulfs of execution and evaluation
 - Identify human errors in the gulfs of execution and evaluation, classify them accordingly
 - Suggest improvements on mitigating these errors